

NOTES MANAGER

Introduction

Notes Manager is a menu-driven Python application that helps users create, organize, update, search, and manage personal or study notes efficiently. The project is built using core Python concepts such as functions, loops, file handling, exception handling, object-oriented programming (OOP), and data analysis with Pandas and NumPy. It also integrates the Groq API with the Llama model to provide AI-powered note summarization and suggestions.

Objectives

- Develop a menu-driven Notes Management System using Python.
- Store and retrieve notes using file handling.
- Apply OOPS concepts with classes and objects.
- Generate basic reports using Pandas and NumPy.
- Implement AI-powered note summarization.
- Handle invalid inputs using exception handling.

Project Overview

The application allows users to add, view, search, update, and delete notes through a simple menu-driven interface. Each note includes a Note ID, Title, Category, Priority, Content, Date, and Status. Notes are stored permanently using file handling, while Pandas and NumPy generate summary reports such as total notes and category-wise analysis. The AI module provides quick summaries and study suggestions. The system is designed to be simple, user-friendly, and suitable for managing both personal and academic notes efficiently.

Technologies Used

Python, Functions, Object-Oriented Programming (OOP), File Handling, Exception Handling, Lists & Dictionaries, CSV, Pandas, NumPy, Groq API, and Llama Model.

Key Features

The application allows users to add, view, search, update, and delete notes through a simple menu-driven interface. It securely stores notes in CSV files, supports password-protected access, validates user inputs using exception handling, and provides category-wise filtering and priority-based sorting. The system also generates basic reports using Pandas and NumPy, while the Groq API with the Llama model offers AI-powered note summarization and study suggestions.

Conclusion

The Notes Manager project demonstrates the practical application of Python by combining file handling, OOP, exception handling, data analysis, and AI integration into a single application. It provides an efficient and user-friendly solution for managing notes while improving programming, analytical, and problem-solving skills.